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DNA color image encryption based on conservative chaotic system

Minxiu Yan, Minghui Liu & Chong Li✉

Simple formal chaotic systems that exhibit complex behaviors continue to garner significant attention. This paper introduces a three-dimensional (3D) conservative chaotic system rooted in generalized Hamiltonian forms. The system's dynamics are comprehensively analyzed employing Lyapunov exponents, phase diagrams, and Poincaré map, demonstrating its capability to transition between periodic and chaotic states. Through alterations in system parameters and initial values, the coexistence of multiple stable states within the system is corroborated. Amplitude modulation control over state variables is achieved by integrating proportionality constants into these variables. Furthermore, the system is applied to a DNA color image encryption algorithm, and its robustness is substantiated through performance analysis and resistance to cropping attacks. This validation underscores the encryption algorithm's excellent confidentiality performance based on the proposed system.

Keywords Conservative chaotic system, Steady-state coexistence, Amplitude modulation control, DNA color image encryption

Since chaos theory was proposed by the United States meteorologist Lorenz[1] in the 1960s, it has gradually evolved into an important branch of science, as evidenced by studies such as[2–5]. With the rapid progress of information technology, the transmission and storage of image data have become increasingly convenient, but this convenience has simultaneously raised serious information security issues. Given the sensitivity of image data and the need for confidentiality, effective protection measures must be implemented during transmission and storage.

Chaotic systems exhibit strong pseudo-randomness and an extreme sensitivity to initial conditions, enabling their integration with various permutation methods for image encryption. Consequently, with the rapid development of the Internet and information technology, image encryption techniques have emerged as a critical focus in information security research[6,7]. For instance, Gao et al.[8] proposed a chaos-based image encryption algorithm, while Xu et al.[9] introduced a novel image encryption scheme based on discrete chaotic systems to enhance data security.

Unlike dissipative chaotic systems, conservative systems lack the typical chaotic attractors observed in dissipative systems, despite sharing the same sensitivity to initial conditions. This distinct feature renders time-delay embedding reconstruction methods ineffective for conservative systems, as their dynamic behaviors become highly unpredictable, thereby significantly enhancing system security[10]. In 2020, Qi Guoyuan et al.[11] proposed a four-dimensional(4D) Euler equation and demonstrated that conservative chaotic systems exhibit superior ergodicity, making them more suitable for applications such as image encryption. Owing to their advantages of ultra-low power consumption, high information storage density, high algorithmic feasibility, and efficient parallelism, DNA coding has been widely adopted in image encryption, particularly in combination with chaotic systems[12,13]. For example, Zhang et al.[14] developed a color image encryption scheme based on a fractional-order 4D hyperchaotic system and DNA sequence operations, incorporating DNA coding rules into the key set and utilizing dual DNA operation principles.

Currently, most chaotic image encryption algorithms rely on dissipative systems[15,16]. To explore the potential of conservative chaotic systems in image encryption, this paper successfully constructs a 3D conservative chaotic system by introducing a generalized Hamiltonian equation. This system not only enables flexible switching between chaotic and periodic states through parameter modulation but also demonstrates multistability coexistence and rich dynamical behaviors. The proposed conservative chaotic system is applied to a DNA-based color image encryption algorithm, significantly improving encryption performance and security. Compared to existing chaotic encryption methods, this algorithm exhibits superior performance in key space size, key sensitivity, and resistance to cropping attacks. Experimental results confirm that the algorithm achieves higher security levels, offering a novel solution in the field of image encryption.

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The main contributions of this paper are as follows:

Theoretical innovation: A 3D conservative chaotic system with multistability coexistence is proposed, expanding the research scope of chaotic dynamics.

Technical breakthrough: A hybrid chaotic image encryption framework is developed. Unlike traditional single-system approaches, this work innovatively integrates conservative chaotic systems, Logistic chaotic systems, and DNA encoding/decoding operations, leveraging their complementary advantages to advance chaos-based encryption.

Reference value: Simulation experiments and performance evaluations provide critical insights for optimizing chaos-driven encryption algorithms.

Application value: The study offers new methodologies for applying chaotic systems to information security and communication encryption.

System construction

The differential equations for the classical conservative chaotic Sprott-A system are as follows:

$$\begin{cases} \dot{x} = y \\ \dot{y} = -x + yz \\ \dot{z} = -y^2 + 2 \end{cases} \tag{1}$$

In a broad sense, the Hamiltonian system is divided into four parts: energy distribution type, energy consumption, energy generation, and energy exchange. The use of matrix differential equations is expressed as:

$$\dot{X} = J(X)\nabla H(X) - \Lambda(X)\nabla H(X) + U \tag{2}$$

In this equation $X = (x, y, z)^{\mathrm{T}}$ is a state variable. *J*(*X*) is an antisymmetric matrix, satisfying $J(X) = J(X)^{\mathrm{T}}$. $\Lambda(X)$ is an indefinite matrix. $\nabla H(X)$ denotes the gradient vector of the Hamiltonian energy *H*(*X*). *U* represents the external torque applied to the system.

According to the generalized Hamiltonian form $\dot{X} = J(X)\nabla H(X) - \Lambda(X)\nabla H(X) + U$, the governing equation of the classical conservative chaotic Sprott-A system is:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & y \\ 0 & -y & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{3}$$

From Eq. (3), we can get the following:

$$J(X) = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & y \\ 0 & -y & 0 \end{bmatrix}, \nabla H(X) = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, U = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{4}$$

It can be seen that the Sprott-A system contains an undissipative vector field associated with energy storage, whereas the vector field represented with energy consumption or generated is 0, it is different from the generalized Hamiltonian form, and the external force parameters of this system can influence the energy conservation state of the system, which can be mathematically described as:

$$\dot{X} = J(X)\nabla H(X) + U \tag{5}$$

Based on the generalized Hamiltonian system framework of the Sprott-A system, we have formulated a system as presented in Eq. (6). This system's structural matrix exhibits distinctive features, notably the presence of zeros along the main diagonal, while the remaining elements exhibit a mirrored pattern across this diagonal, effectively reversing their positions with respect to the main axis.

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 0 & a & cx \\ -a & 0 & b(x+y) \\ -cx & -b(x+y) & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{6}$$

From Eq. (6), the system equation can be obtained:

$$\begin{cases} \dot{x} = ay - cxz \\ \dot{y} = -ax - bz(x+y) \\ \dot{z} = cx^2 - by(x+y) + 1 \end{cases} \tag{7}$$

Where *x*, *y*, and *z* are state variables, and *a*, *b* and *c* are the parameters of the new system. When the parameters $a = 1, b = 1.9, c = 1$, and the initial values of the system are $(x_0, y_0, z_0) = (0, 1, 1)$, the phase trajectories of the system are shown in the Fig. 1.

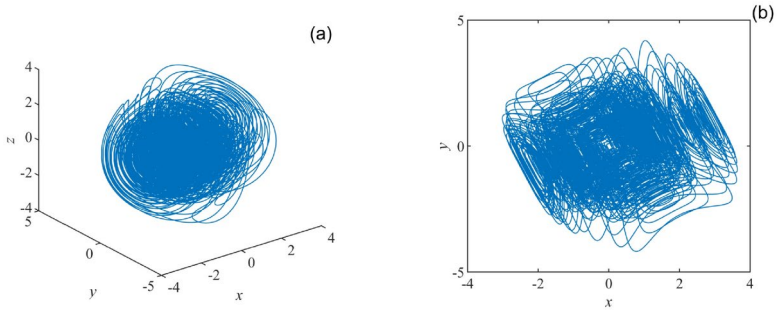


Fig. 1. Phase trajectories with fixed parameters $a = 1, b = 1.9, c = 1$ and initial conditions of $(x_0, y_0, z_0) = (0, 1, 1)$. (**a**) $x - y - z$ 3D space. (**b**) $x - y$ plane..

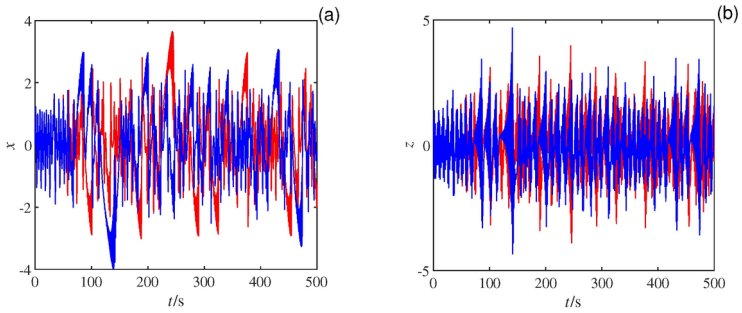


Fig. 2. Time series diagrams with the initial values $(x_0, y_0, z_0) = (0, 1, 1)$ represented by red and $(x_0, y_0, z_0) = (0.001, 1, 1)$ represented by blue. (**a**) Comparison of x time series diagrams. (**b**) Comparison of z time series diagrams.

Analysis of dynamic characteristics

Initial values sensitivity

Chaotic systems are highly sensitive to initial conditions and are analyzed for the sensitivity of the new system to these initial values. The system parameters are selected as $a = 1$, $b = 1.9$, and $c = 1$. With initial conditions of $(x_0, y_0, z_0) = (0, 1, 1)$ and $(x_0, y_0, z_0) = (0.001, 1, 1)$, respectively, a comparison of the time series diagrams for the state variables x and z under different initial conditions is presented in Fig. 2. The red time series represents the initial values $(x_0, y_0, z_0) = (0, 1, 1)$, the blue time series represents the initial values $(x_0, y_0, z_0) = (0.001, 1, 1)$. Fig. 2a shows that the trajectory of the x time series completely separates at approximately 49s and undergoes significant changes. As demonstrated in Fig. 2b, the z time series further separates into distinct trajectories around this time point (49s). These results indicate that the new system exhibits high sensitivity to initial values.

Dissipation

The dissipation degree of system (7) is:

$$\nabla V = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z}$$

The divergence calculation shows that $\nabla V = (b - c)z$. When $b = c$, the system volume is conserved. However, this paper selects volume as a system parameter under the condition $a \neq b$. Therefore, even when $z = 0$ and the system dissipation vanishes ($\nabla V = 0$), system (7) cannot be guaranteed to exhibit chaotic behavior with volume conservation.

Lyapunov exponents and fractal dimensions

Lyapunov exponents are the critical tool for analyzing the dynamics of chaotic systems. They allow us to determine the system's state. For parameters $a = 1, b = 1.9, c = 1$ and initial conditions $(x_0, y_0, z_0) = (0, 1, 1)$, the Lyapunov exponents of the proposed chaotic system are shown in Fig. 3, where $LE_1 = 0.20$, $LE_2 = 0$, and $LE_3 = -0.20$. The positive value of LE_1 confirms that the system exhibits chaotic behavior. At the same time, the sum of these three Lyapunov exponents is obtained as $LE_s = 0$, which indicates that a chaotic system can exhibit conservative properties.

Lyapunov fractal dimensionality D_L is also an important indicator to measure whether a system is chaotic or not. Fractal dimensionality D_L can be calculated according to the Lyapunov exponents, as shown in Eq. (8):

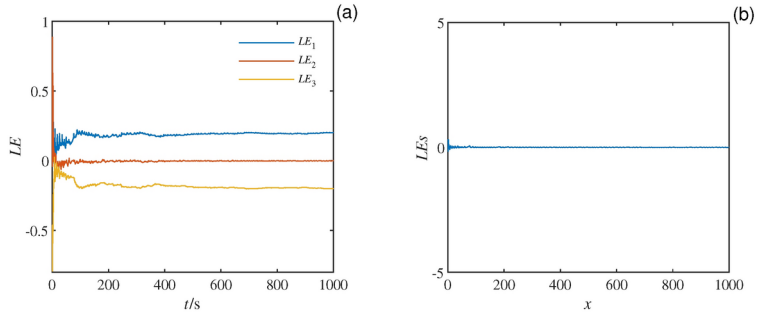


Fig. 3. Lyapunov exponents analysis under fixed parameters. (**a**) Lyapunov exponents. (**b**) Sum of the Lyapunov exponents.

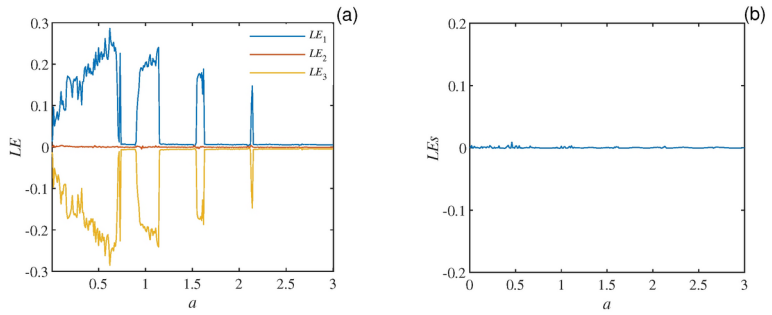


Fig. 4. Lyapunov exponents analysis with variable parameter *a*. (**a**) Lyapunov exponents diagram with parameter *a*. (**b**) Sum of the Lyapunov exponents.

$$D_L = j + \frac{1}{|LE_{j+1}|}\sum_{i=1}^{j} LE_i \tag{8}$$

where j denotes the maximum integer satisfying both conditions $\sum_{i=1}^{j} LE_i \geq 0$ and $\sum_{i=1}^{j+1} LE_i \leq 0$. For $j = 2$, the Lyapunov dimension is expressed as: $D_L = 2 + \frac{1}{|LE_3|}|LE_1 + LE_2|$ Substituting the specific numerical values: $D_L = 2 + \frac{1}{|-0.20|}|0.20 + 0| = 2 + 1 = 3$. This result demonstrates that the Lyapunov dimension of the system is an integer equal to the system's dimensionality, thereby confirming that the system belongs to a chaotic system.

Effect of parameter changes on the system

The system (7) contains three variable parameters with unknown specific values, namely the linear term parameter *a*, and the nonlinear term parameters *b* and *c*. To investigate the influence of these parameters on the system's conservatism and chaotic behavior, this section employs multiple research methods to quantitatively analyze the dynamic characteristics of the system under varying parametric conditions.

Effect of parameter a on the system

The first is the analysis of the linear term parameter *a*. The initial values of the system are (0, 1, 1), with parameters $b = 1.9$ and $c = 1$. For *a* in the range [0, 3], the Lyapunov exponents are plotted as a function of *a*. An in-depth analysis of Fig. 4a reveals that as *a* varies from 0 to 3, the system's state undergoes significant transformations in complexity. Notably, the maximum Lyapunov exponent exhibits considerable fluctuations, indicating the dynamic nature and potential transitions in the system's behavior. When $a \in [0, 0.73] \cup [0.94, 1.14] \cup [1.55, 1.62] \cup [2.12, 2.15]$, the maximum Lyapunov exponent is greater than 0, and the system is in a chaotic state; when $a \in [0.73, 0.94] \cup [1.14, 1.55] \cup [1.62, 2.12] \cup [2.15, 3]$, the maximum Lyapunov exponent is not greater than 0, and the system is in a periodic state. This demonstrates that the system transitions between chaotic and periodic states as *a* varies, despite fixed initial conditions. From Fig. 4b, the Lyapunov exponents diagram oscillates near zero with fluctuating magnitudes, suggesting the system does not always maintain a conservative chaotic state.

For quantitative evaluation, several representative systems with distinct values of the parameter *a* are meticulously selected. As an initial step, the specific value of $a = 0.5$ is chosen, and the corresponding $x - y - z$ phase trajectories are presented in Fig. 5a. This spatial phase diagram vividly depicts the chaotic topology inherent in the system. Furthermore, Fig. 5b displays the Poincaré map with *z* as the cross-sectional

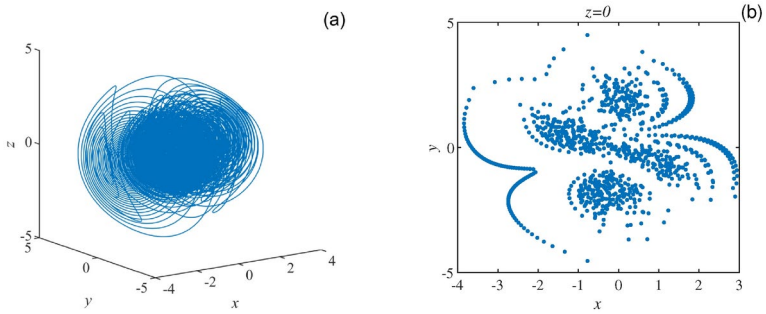


Fig. 5. Phase trajectories and Poincaré map of the system with variable parameter $a = 0.5$. (**a**) $x - y - z$ 3D space. (**b**) Poincaré map at $z = 0$.

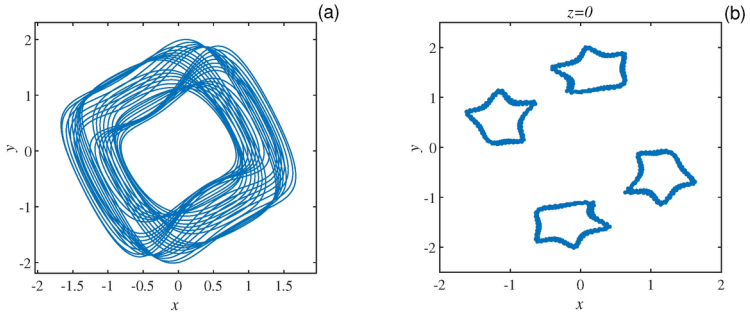


Fig. 6. Phase diagram and Poincaré map of the system with variable parameter $a = 1.5$. (**a**) $x - y$ plane. (**b**) Poincaré map at $z = 0$.

plane, revealing an irregular ensemble of points characterized by dense clustering. At this parameter setting, the system exhibits a mix of positive, zero, and negative Lyapunov exponents, conclusively indicating chaotic state dynamics. The sum of the Lyapunov exponents equals zero, affirming the conservative nature of the system.

When $a = 1.5$, the phase diagram in the $x - y$ plane is shown in Fig. 6a, where the spatial topology exhibits periodicity. The Poincaré map with z as the cross-section (Fig. 6b) displays a closed curve formed by densely distributed points, indicating a periodic state of the system. The sum of the Lyapunov exponents equals zero, which is consistent with the conservative nature of the system.

From the above conclusions, it can be concluded that the change of the linear term parameter *a* can affect the state of the system, so that the system changes between conservative chaos and conservative periods.

Effect of parameter b on the system

Subsequently, we delve into analyzing parameter *b* associated with the nonlinear term. For this investigation, the initial values of the system are set to (0, 1, 1), while parameters *a* and *c* are fixed at -1 and 1, respectively. Within the range $b \in [1, 3]$, we plot the Lyapunov exponents as a function of *b* to explore its variation with respect to this crucial parameter.

Analysis of Fig. 7a reveals a continuous interval of system states as parameter *b* varies from 1 to 3. Notably, except at $b = 1$, the system consistently exhibits positive Lyapunov exponents, confirming its chaotic nature within this range. In Fig. 7b, we observe that both the Lyapunov exponents and its corresponding curve approximate zero, despite fluctuations in specific values. This observation indicates that the system does not maintain a conservative chaotic state uniformly, suggesting potential variations in its dynamical behavior.

Several typical systems with different values of parameter *b* are selected for quantitative analysis. First, we consider the case of $b = 1$. The $x - y - z$ phase trajectories are shown in Fig. 8a, and the corresponding Poincaré map with z as the cross-section is presented in Fig. 8b, revealing independent points. The system exhibits two Lyapunov exponents equal to zero, indicating a quasi-periodic state. Furthermore, the sum of all Lyapunov exponents is zero, confirming the conservative nature of the system.

The parameter $b = 2$ is selected. The $x - y - z$ phase trajectories are shown in Fig. 9a, while the spatial phase trajectories present a chaotic topology. The Poincaré map with z as the cross-section is illustrated in Fig. 9b, represented by a dense set of points. Under this parameter, the system exhibits a positive Lyapunov exponent, indicating a chaotic state. Additionally, the sum of the Lyapunov exponents is 0, confirming the system's conservative nature.

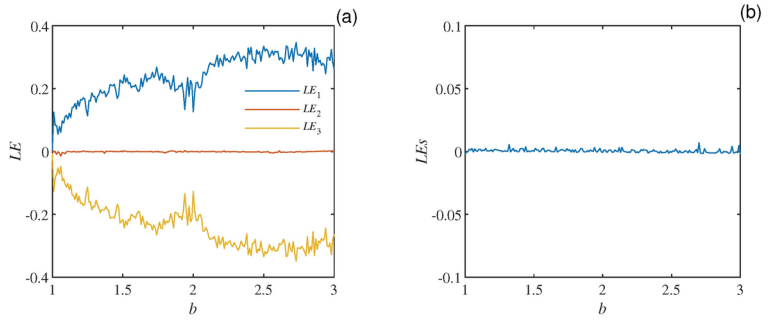


Fig. 7. Lyapunov exponents analysis with variable parameter b. (**a**) Lyapunov exponents diagram with parameter b. (**b**) Sum of the Lyapunov exponents.

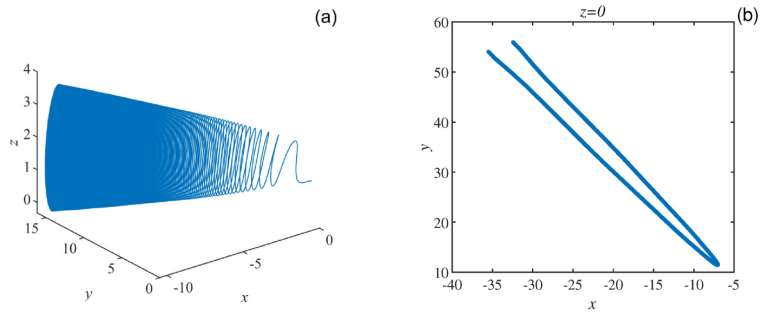


Fig. 8. Phase trajectories and Poincaré map of the system with variable parameter $b = 1$. (**a**) $x - y - z$ 3D space.(**b**) Poincaré map at $z = 0$.

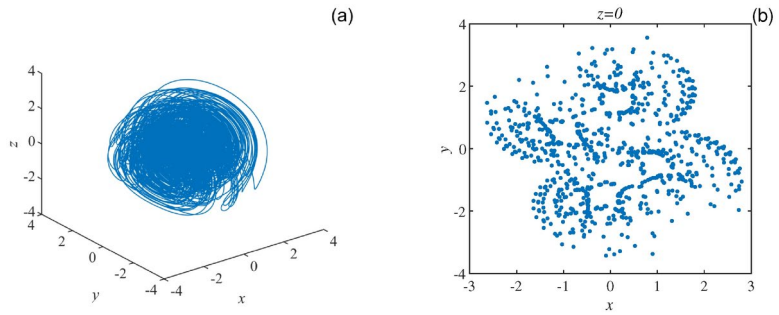


Fig. 9. Phase trajectories and Poincaré map of the system with variable parameter $b = 2$. (**a**) $x - y - z$ 3D space. (**b**) Poincaré map at $z = 0$.

Effect of parameter c on the system

We then analyze the nonlinear parameter c, with the system initialized at (0, 1, 1) and fixed parameters $a = 1$ and $b = 1.9$. For $c \in [-1.5, 1.5]$, the Lyapunov exponents are plotted as a function of c.

Fig. 10a reveals that as c varies within this range, the system undergoes complex dynamical transitions, accompanied by significant fluctuations in the maximum Lyapunov exponent. When $c \in [-1.5, 0.15] \cup [0.47, 0.76] \cup [0.95, 1.1] \cup [1.19, 1.5]$, the maximum Lyapunov exponent is positive, indicating a chaotic state. Conversely, for $c \in [0.15, 0.47] \cup [0.76, 0.95] \cup [1.1, 1.19]$, the maximum Lyapunov exponent is non-positive, corresponding to periodic dynamics. These results demonstrate that the system alternates between chaos and periodicity as parameter c varies, given a fixed initial state. As shown in Fig. 10b, the Lyapunov exponents curve oscillates near zero with fluctuating amplitudes. This behavior further confirms that the system does not always maintain a conservative chaotic state.

Several typical systems with different values of parameter c are selected for quantitative analysis. First, the value $c = -1$ is chosen. The corresponding $x - y$ phase diagram is shown in Fig. 11a, exhibiting a chaotic topology. The Poincaré map with z as the cross-sectional plane is presented in Fig. 11b, revealing an irregular set

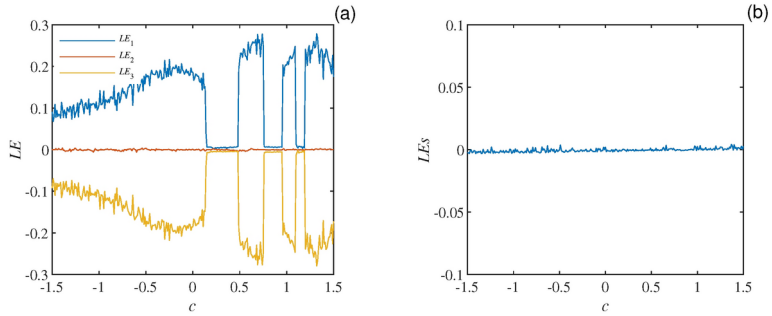


Fig. 10. Lyapunov exponents analysis with variable parameter c. (**a**) Lyapunov exponents diagram with parameter c. (**b**) Sum of the Lyapunov exponents.

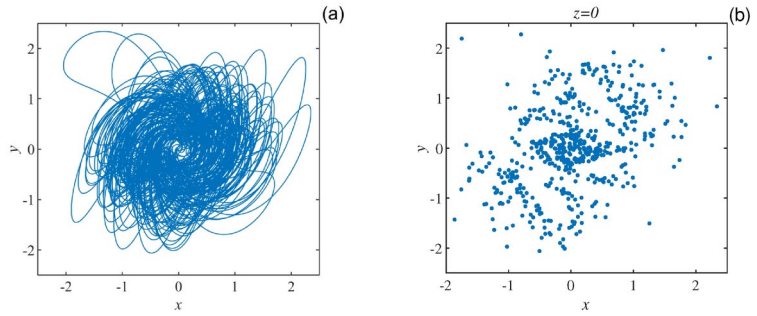


Fig. 11. Phase diagram and Poincaré map of the system with variable parameter $c = -1$. (**a**) $x - y$ plane. (**b**) Poincaré map at $z = 0$.

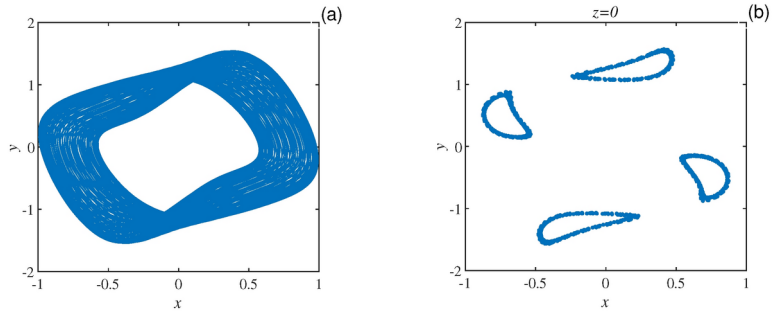


Fig. 12. Phase diagram and Poincaré map of the system with variable parameter $c = 0.3$. (**a**) $x - y$ plane. (**b**) Poincaré map at $z = 0$.

of points composed of dense clusters. In this case, the system possesses a positive largest Lyapunov exponent, indicating chaotic behavior.

When parameter c is set to 0.3, the $x - y$ phase diagram is shown in Fig. 12a, while the Poincaré map with z as the cross-sectional plane is illustrated in Fig. 12b, revealing a closed figure formed by dense points. Under this condition, the system exhibits no positive Lyapunov exponents and remains in a periodic state. Notably, the sum of Lyapunov exponents equals zero, indicating that the system neither expands nor contracts exponentially, consistent with its periodic behavior characteristic of conservative dynamics. Furthermore, the system is conservative, as no net energy dissipation or gain occurs over time.

These observations demonstrate that variations in parameters a (linear term), b (nonlinear term), and c significantly alter the system's state, inducing transitions between periodic and chaotic regimes. Such rich dynamical behavior may enhance image encryption algorithms by leveraging chaotic states for higher security.

Amplitude modulation control of chaotic systems

When simulating chaotic circuits, it is necessary to implement linear compression of the dynamic range of the system state variables to ensure that the operational amplifiers (op amps) and components operate within the

safe voltage domain. For chaotic system applications, the signal amplitude needs to be flexible. Therefore, it is essential to develop a technique that can adjust the amplitude of chaotic signals without disrupting the state and performance of the system.

To control the amplitude of the state variables in system (7) and thereby modulate the amplitude of the generated chaotic signal, the state variables can be scaled by introducing proportionality constants k_1x, k_2y, and k_3z. Specifically, the original state variables x, y, and z are replaced with k_1x, k_2y, and k_3z, where k_1, k_2, and k_3 are amplitude control constants. Consequently, system (7) can be expressed as:

$$\begin{cases} \dot{x} = (ak_2y - ck_1k_3xz)/k_1 \\ \dot{y} = [-ak_1x + bk_3z(k_1x + k_2y)]/k_2 \\ \dot{z} = [ck_1^2x^2 - bk_2y(k_1x + k_2y) + 1]/k_3 \end{cases} \tag{9}$$

The initial values of system (9) are set to (1, 1, 1), with parameters $a = 1, b = 1.9$ and $c = 1$. When the amplitude modulation parameters are set to $k_1 = 1, k_2 = 1, k_3 = 1$, Fig. 13a shows the $x - z$ phase diagram of the system, Fig. 13b presents the Lyapunov exponents diagram, and Fig. 13c and d illustrate the x and z time series diagrams, providing a time-domain view of the dynamic behavior. The Hamiltonian energy of the system is:
$H(X) = \frac{1}{2}(x^2 + y^2 + z^2) = \frac{3}{2}$.

Amplitude modulation of single-state variables

The amplitude control of a single signal amplifies the signal strength of x by a factor of 100, with the initial conditions set to (100, 1, 1). Consequently, the amplitude modulation parameters are configured as $k_1 = 0.01$, $k_2 = 1$, and $k_3 = 1$. The system's $x - z$ phase portrait is depicted in Fig. 14a, while the corresponding Lyapunov exponents diagram, x-time series, and z-time series are presented in Fig. 14b–d respectively.

Its Hamiltonian energy is $H(X) = \frac{1}{2}(x^2 + y^2 + z^2) = \frac{3}{2}$.

After amplitude modulation of a single signal, the x-component amplitude increases by a factor of 100. This scaling is visualized in both the $x - z$ phase diagram and x time series diagram. Calculation of the Lyapunov exponents (Fig. 14b) confirms the system maintains chaotic behavior. Notably, the Hamiltonian energy remains unchanged throughout the modulation process, demonstrating system property conservation.

Amplitude modulation of multi-state variables

For multiple signal control, simultaneous scaling of x ($\times 100$) and z ($\times 0.1$) amplitudes yields initial conditions (100, 1, 0.1), with modulation parameters set as $k_1 = 0.01$, $k_2 = 1$, and $k_3 = 10$. The resulting dynamics are characterized by: (a) $x - z$ phase diagram, (b) Lyapunov exponents diagram, (c) x time series, and (d) z time series diagrams (Fig. 15).

Its Hamiltonian energy is $H(X) = \frac{1}{2}(x^2 + y^2 + z^2) = \frac{3}{2}$.

After multi-signal amplitude modulation of the system, the x-signal amplitude exhibits a 100-fold increase, while the z-signal amplitude decreases by a factor of 10. These changes are evident in both the phase diagram

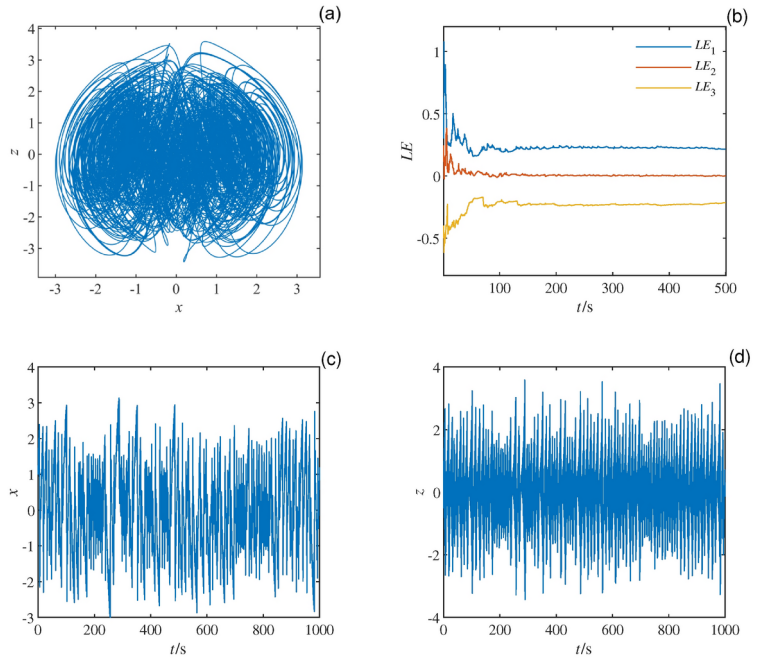


Fig. 13. Phase diagram, Lyapunov exponents diagram and time series diagrams before amplitude modulation. (**a**) $x - z$ phase diagram. (**b**) Lyapunov exponents diagram. (**c**) x time series diagram. (d) z time series diagram.

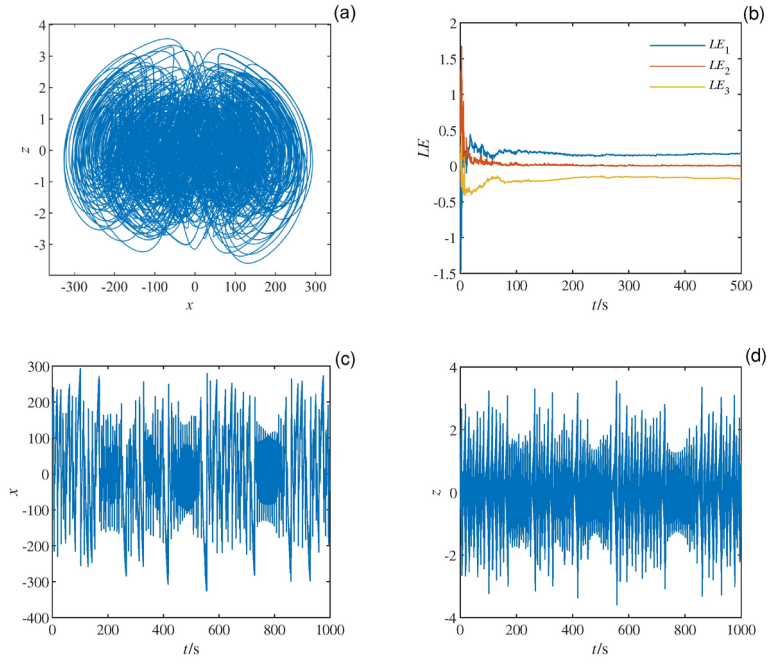


Fig. 14. Phase diagram, Lyapunov exponents diagram and time series diagrams of the system after single signal amplitude modulation. (**a**) $x - z$ phase diagram. (**b**) Lyapunov exponents diagram. (**c**) x time series diagram. (**d**) z time series diagram.

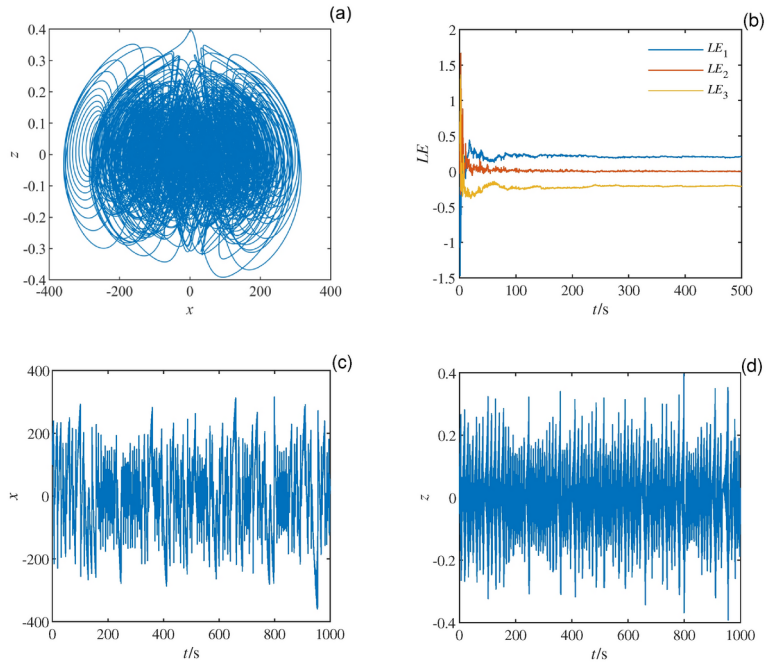


Fig. 15. Phase diagram, Lyapunov exponents diagram and time series diagrams of the system after amplitude modulation of multiple signals. (**a**) $x - z$ phase diagram. (**b**) Lyapunov exponents diagram. (**c**) x time series diagram. (**d**) z time series diagram.

and the x and z time series diagrams. Calculation of the Lyapunov exponents confirms that the system remains in a chaotic state. Notably, no variation is observed in the Hamiltonian energy, demonstrating that the system's fundamental properties are conserved.

Steady-state coexistence of chaotic systems

Unlike dissipative chaotic systems, conservative chaotic systems do not possess attractors. However, their chaotic trajectories exhibit topological structures resembling attractors, known as steady-state chaotic flow clusters. A special coexistence phenomenon observed in some dissipative systems involves multiple attractors emerging from different initial values under identical parameters. In conservative systems, this multistability phenomenon is termed homeostatic coexistence.

When the parameters are set to $a = 1$, $b = 1.9$, $c = 1$, with initial values (0, 1, 1) and $(0, -1, -1)$, the corresponding $x - y$ phase diagrams and (p, s) motion trajectories are plotted in Fig. 16. In this figure, red curves represent trajectories from the initial values (0, 1, 1), while blue curves correspond to $(0, -1, -1)$. The Lyapunov exponents calculated for Fig. 16a are $LE_1 = 0.20$, $LE_2 = 0$, $LE_3 = -0.20$, confirming chaotic behavior for the red initial conditions. Similarly, the exponents in Fig. 16b $LE_1 = 0.22, LE_2 = 0, LE_3 = -0.22$ indicate chaos for the blue initial values. The irregular trajectories shown in Fig. 16c and d exhibit characteristics consistent with chaotic systems, further evidencing the multistable coexistence within this dynamical system.

When the parameters are set to $a = 1, b = 1.9, c = 0.3$ with initial values (0, 1, 1) and $(0, -1, -1)$, the $x - y$ phase diagrams and (p, s) motion trajectory diagrams are presented in Fig. 17. The red trajectory corresponds to initial values (0, 1, 1), while the blue trajectory represents $(0, -1, -1)$. The Lyapunov exponents calculated for Fig. 17a are $LE_1 = 0$, $LE_2 = 0$, $LE_3 = 0$, indicating a quasi-periodic state for the red initial conditions, as shown in Fig. 17c. In contrast, the exponents for Fig. 17b are $LE_1 = 0.17$, $LE_2 = 0$, $LE_3 = -0.17$, demonstrating chaotic behavior for the blue initial conditions. This chaotic state is consistent with the irregular motion observed in Fig. 17d. These results collectively suggest multistable coexistence in the system.

DNA color image encryption based on chaotic systems

The proposed encryption algorithm comprises three main phases. First, the color image is decomposed into three 2D matrices corresponding to the R, G, and B channels, each of which is subsequently partitioned into sub-blocks. Second, pseudo-random sequences are generated through a novel 3D conservative chaotic system combined with a 1D Logistic chaotic mapping. These sequences govern both the DNA encoding and algebraic operations performed on the partitioned sub-blocks. Finally, the encoded DNA matrices are decoded using the pseudo-random sequences, followed by row-column permutation. The ciphertext image is then reconstructed by merging the processed matrices. The complete encryption workflow is illustrated in Fig. 18.

Encryption algorithm steps

Step 1: For an $M \times N$ image to be encrypted, separate the R, G, and B channels into three 2D matrices C_1, C_2, C_3. Each matrix is divided into blocks, with any missing portions padded with zeros.

Step 2: Set the system parameters μ and initial values x, x_1, x_2 of the logistic mapping; generate pseudorandom sequences $\{Q_i\}$, $\{S_i\}$, $\{T_i\}$. Derive x, x_1, x_2 from the following equation:

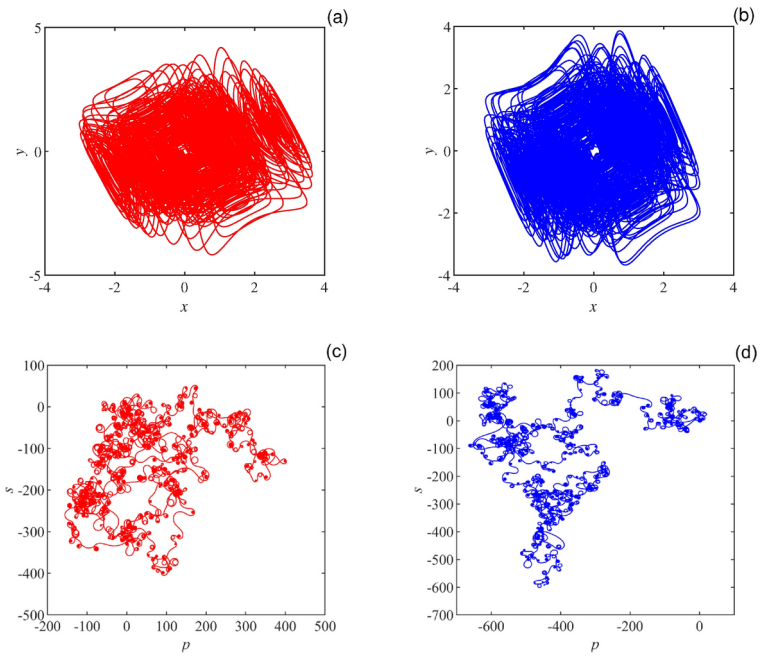


Fig. 16. Phase diagrams and $0-1$ test diagrams. (**a**) Initial values (0, 1, 1). (**b**) Initial values $(0, -1, -1)$. (**c**) (0, 1, 1) (p, s) trajectory diagram. (**d**) $(0, -1, -1)$ (p, s) trajectory diagram.

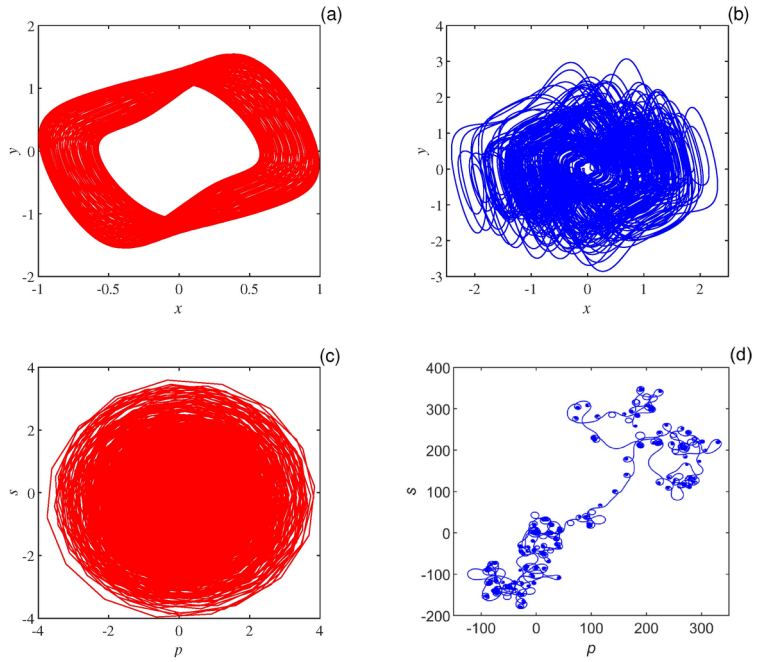


Fig. 17. Phase diagrams and $0-1$ test diagrams. (**a**) Initial values (0, 1, 1). (**b**) Initial values $(0,-1,-1)$. (**c**) (0, 1, 1) (p, s) trajectory diagram. (**d**) $(0,-1,-1)$ (p, s) trajectory diagram.

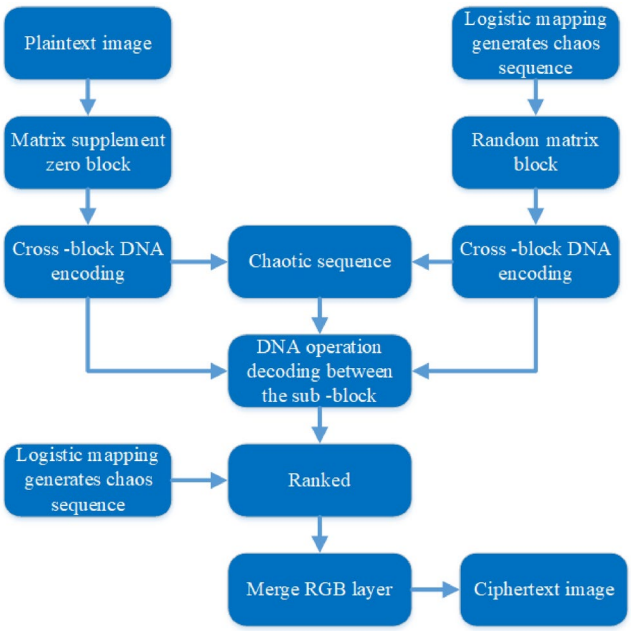


Fig. 18. Encryption flow chart.

$$\begin{cases} x = \mathrm{sum}(C_1(:)) + \mathrm{sum}(C_2(:))/255 \times M \times N \times 2 \\ x_1 = \mathrm{sum}(C_2(:)) + \mathrm{sum}(C_2(:))/255 \times M \times N \times 2 \\ x_2 = \mathrm{sum}(C_3(:)) + \mathrm{sum}(C_3(:))/255 \times M \times N \times 2 \end{cases} \tag{10}$$

Step 3: Convert the pseudo-random sequence $\{Q_i\}$ into an $M \times N$ matrix with values in the range 0-255 for DNA computation with $C_i(i = 1, 2, 3)$.

$$\begin{cases} Q = \mathrm{mod}(round(k \times 10^4), 256) \\ A = \mathrm{reshape}(Q, N, M)' \end{cases} \tag{11}$$

Step 4: Set the initial values x_0, y_0, z_0 of the new 3D conservative chaotic system. These initial values are calculated by the following equation to generate the pseudo-random sequences $\{X_i\}$, $\{Y_i\}$, $\{Z_i\}$.

$$\begin{cases} x_0 = sum(sum(bitand(C_1, 17)))/(17 \times M \times N) \\ y_0 = sum(sum(bitand(C_2, 34)))/(34 \times M \times N) \\ z_0 = sum(sum(bitand(C_3, 68)))/(68 \times M \times N) \end{cases} \tag{12}$$

Step 5: Specify the encoding method; the subblocks in the same position within C_1, C_2, and C_3 use the same encoding method determined by $\{X_i\}$. Each subblock of matrix *A* is determined by $\{Y_i\}$. Since there are 8 possible DNA encoding methods, the values of sequences $\{X_i\}$ and $\{Y_i\}$ must be converted into integers ranging from 1 to 8. This conversion is implemented using the following equation:

$$\begin{cases} X = \mathrm{mod}(round(X \times 10^4), 8) + 1 \\ Y = \mathrm{mod}(round(Y \times 10^4), 8) + 1 \end{cases} \tag{13}$$

Step 6: Specify the operation rules of each sub-block. The same algorithm is used among C_1, C_2, C_3, and the corresponding block of the matrix *A*, as determined by $\{Z_i\}$. Subsequently, the current sub-block and the previous sub-block in the calculated matrix are combined with each other, which is also governed by $\{Z_i\}$. Therefore, the sequence $\{Z_i\}$ needs to be transformed into integers ranging from 0 to 3 according to the following equation:

$$Z = \mathrm{mod}(round(Z \times 10^4), 4) \tag{14}$$

The rule states: if $Z_i = 0$, use addition operation; if $Z_i = 1$, use subtraction operation; if $Z_i = 2$, use XOR operation; if $Z_i = 3$, use XNOR operation.

Step 7: Generate the pseudo-random sequence $\{W_i\}$ and define the decoding rules for each sub-block.

$$\begin{cases} w_0 = sum(sum(bitand(C_1, 136)))/(136 \times M \times N) \\ W = \mathrm{mod}(round(W \times 10^4), 8) + 1 \end{cases} \tag{15}$$

Step 8: Row and Column Permutation. The pseudo-random sequences $\{S_i\}$ and $\{T_i\}$ are sorted in descending order to obtain the original position sequences U_i and V_i. These sequences are then utilized to permute the rows and columns of the three-channel matrix, enhancing the scrambling effect. Additionally, this step strengthens the ciphertext image's resistance to clipping attacks.

$$\begin{cases} [\sim, U_i] = sort(S_i, 'descend') \\ [\sim, V_i] = sort(T_i, 'descend') \end{cases} \tag{16}$$

Step 9: Combine the three 2D matrices (used for row and column substitution) into a 3D matrix, thereby generating the ciphertext image.

Decryption algorithm steps

The decryption step is the reverse process of the encryption step. First, the ciphertext image is transformed into rows and columns. Next, the DNA sequences are divided into blocks for DNA coding, operation phase, and decoding. This step specifically involves interchanging the DNA addition and subtraction operations to ensure decryption correctness. Finally, the padded zeros are removed, the three separate 2D matrices are merged into a single 3D matrix, and the decrypted image is output.

Experimental results and performance analysis

Encryption and decryption

Three color images ("Longmen", "Flower", and "Orange"), each with dimensions of 512×512 pixels, are selected for encryption. The key generation algorithm, image processing functions, and encryption simulation scripts are implemented in MATLAB R2019a. Notably, the encryption key *K* in this algorithm is dynamically generated based on the pixel characteristics of the input color image, which ensures distinct keys for each of the three test images. Specifically, the following implementation details should be emphasized:

- $K_1 = [3.9999, 0.3794, 0.5551, 0.5701, 0.4430, 0.1841, 0.4275, 0.5242]$
- $K_2 = [3.9999, 0.5664, 0.5190, 0.3827, 0.4953, 0.3774, 0.3718, 0.6860]$

- $K_3 = [3.9999, 0.5867, 0.4191, 0.1032, 0.6005, 0.4699, 0.1163, 0.8235]$

Fig. 19 demonstrates the encryption and decryption effects of the proposed algorithm. From left to right, the subfigures represent the plaintext, ciphertext, and decrypted image, respectively. As shown in Fig. 19, the ciphertext exhibits no visible correlation with the original image after encryption. Moreover, the decrypted image perfectly reconstructs the original content without information loss. These results confirm the excellent encryption performance of the algorithm.

Histogram

Histograms are an important tool for analyzing image information, as they display the distribution of pixel values to evaluate encryption effectiveness. The more uniform the pixel distribution, the weaker the statistical correlation between pixel values and their intensity values. When the encrypted image is attacked, adversaries can obtain less meaningful information. In Fig. 20, the left histograms show the RGB channel distributions (blue for R, red for G, yellow for B) of the original images, while the right histograms correspond to the ciphertext images after encryption. The uniform distributions observed in the encrypted image's channel histograms demonstrate the algorithm's strong encryption performance.

Key analysis

In the algorithm proposed in this paper, the key K comprises eight components: μ, x, x_0, y_0, z_0, w_0, x_1 and x_2. By adjusting the decimal precision of each key component during image decryption, the key sensitivities for μ, x, x_0, y_0, z_0, w_0, x_1 and x_2 are determined to be 10^{-15}, 10^{-16}, 10^{-16}, 10^{-16}, 10^{-16}, 10^{-16}, 10^{-16} and 10^{-16} respectively. Consequently, the total secret key space is calculated as: $10^{-15} \times 10^{-16} \times 10^{-16} \times 10^{-16} \times 10^{-16} \times 10^{-16} \times 10^{-16} \times 10^{-16} = 10^{127} \approx 2^{422}$ Given that the key space of an ideal image encryption algorithm should exceed 2^{122}, our result demonstrates significant robustness. Furthermore, the key capacity can be extended by configuring distinct μ values in the logistic chaotic system, indicating expandable security margins.

As shown in Table 1, the comparative analysis with existing encryption algorithms confirms that our key space is sufficiently large to resist exhaustive key search attacks and effectively safeguard encrypted images.

Key sensitivity refers to the impact of minor key modifications during the encryption and decryption processes on the resulting decrypted image. A slight key alteration can visually demonstrate the algorithm's sensitivity to key variations.

Taking the Longmen image as an example, we first encrypt the image Fig. 21a. When decrypting with a fixed key where μ is perturbed by 10^{-10}, the result is Fig. 21b. In contrast, decryption with the correct key K yields

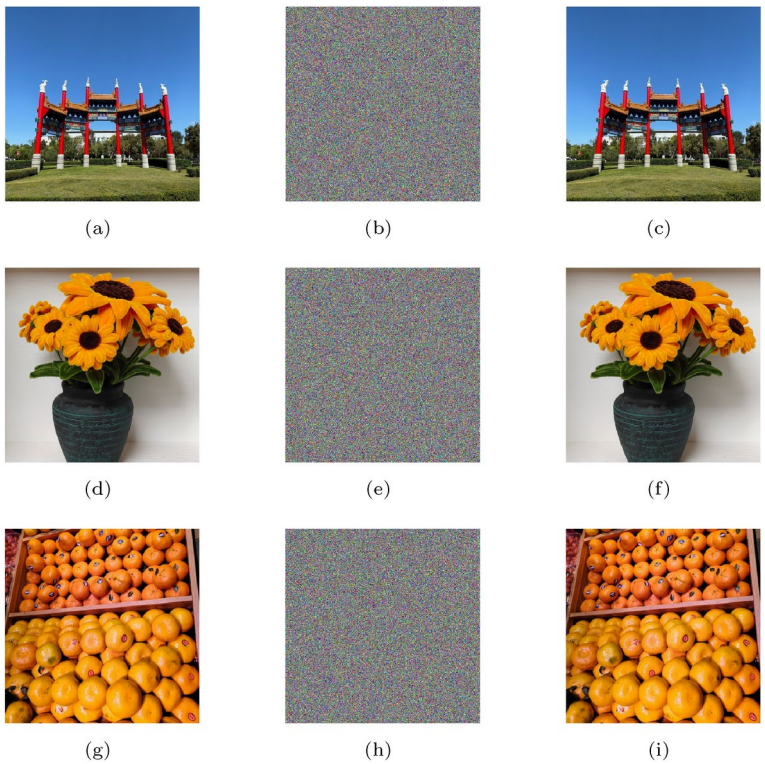


Fig. 19. Image encryption and decryption. (**a**) Plaintext images of Longmen. (**b**) Redaction images of Longmen. (**c**) Decrypt the image of Longmen. (**d**) Plaintext images of Flower. (**e**) Redaction images of Flower. (**f**) Decrypt the image of Flower. (**g**) Plaintext images of Orange. (**h**) Redaction images of Orange. (**i**) Decrypt the image of Orange.

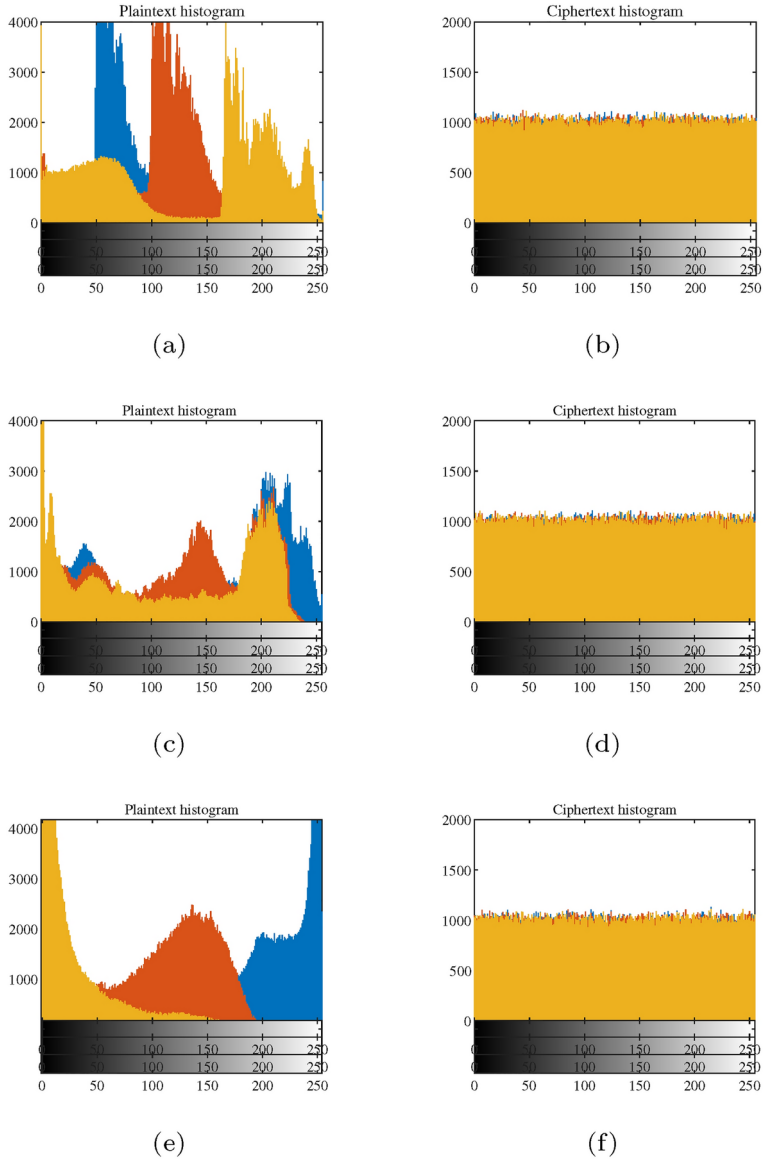


Fig. 20. Histogram. (**a**) Plaintext histogram of Longmen. (**b**) Ciphertext histogram of Longmen. (**c**) Plaintext histogram of Flower. (**d**) Ciphertext histogram of Flower. (**e**) Plaintext histogram of Orange. (**f**) Ciphertext histogram of Orange.

Algorithm	This paper	Ref. [17]	Ref. [18]	Ref. [19]
Key space	$10^{127} \approx 2^{422}$	$10^{79} \approx 2^{262}$	$10^{90} \approx 2^{299}$	$10^{56} \approx 2^{168}$

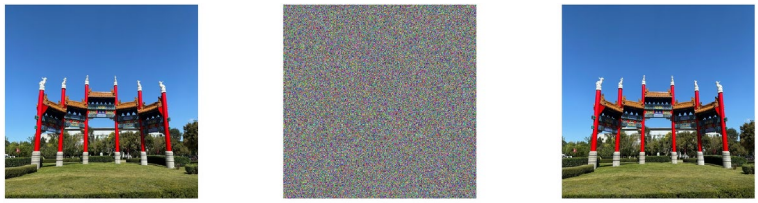
Table 1. Key space comparison results.

Fig. 21c. As shown in Fig. 21, even a slight key modification renders the decrypted image entirely unrelated to the original plaintext, demonstrating the algorithm's high key sensitivity.

Information entropy

The greater the information entropy, the higher the degree of disorder in the system, and the less discernible the information becomes. The equation for information entropy is defined as follows:

$$H = -\sum_{i=0}^{L} p(i) \log_2 p(i) \tag{17}$$



(a) (b) (c)

Fig. 21. Key sensitivity. (**a**) Original image. (**b**) μ error. (**c**) Clealing image.

Image	Channel	Information entropy (plaintext/ciphertext)
Longmen	R-channel	7.1547/7.9992
	G-channel	7.2758/7.9993
	B-channel	7.5325/7.9993
Flower	R-channel	7.6231/7.9993
	G-channel	7.7066/7.9993
	B-channel	6.9718/7.9992
Orange	R-channel	7.2171/7.9992
	G-channel	7.4920/7.9993
	B-channel	5.8266/7.9993

Table 2. Comparison of information entropy between raw and ciphertext images.

Channel	This paper	Ref. [20]	Ref.[21]	Ref.[22]
R-channel	7.9992	7.9980	7.9958	7.9991
G-channel	7.9993	7.9979	7.9950	7.9993
B-channel	7.9993	7.9978	7.9949	7.993

Table 3. Comparison of information entropy.

where L denotes the gray level (256 for each RGB channel in this study) and $p(i)$ represents the occurrence probability of gray value i. The theoretical maximum entropy for 8-bit images is 8, with higher values indicating greater randomness. We evaluate the encryption algorithm using three test images ("Longmen", "Flower", and "Orange"). Table 2 presents the entropy measurements for both plaintext and ciphertext images across RGB channels. The ciphertext entropy values approach the theoretical maximum of 8 in all three channels, demonstrating that the encrypted images exhibit near-optimal randomness. This suggests the proposed algorithm effectively resists information entropy-based cryptanalysis.

As shown in Table 3, the information entropy calculated by our algorithm (7.9993) approaches the theoretical limit more closely than values reported in existing literature, indicating its advantages.

Anti-cropping performance

When subjected to cropping attacks, the ciphertext image should maintain minimal information loss in the decrypted version while preserving overall integrity. In this experiment, the Longmen image is selected as the plaintext for encryption. We implement cropping attacks on the ciphertext by removing central regions accounting for 10 %, 20 %, and 30 % of the original image area, with the cropped pixels set to 255. The decryption results of the tampered ciphertext are presented in Fig. 22. Although partial clipping damages the ciphertext image, the proposed algorithm effectively propagates the influence of cropped regions across the entire image, thereby significantly mitigating localized distortion. These results demonstrate the algorithm's robust resistance to cropping attacks.

Correlation analysis

For digital images, the larger the value of their adjacent pixel correlation coefficient, the higher the degree of adjacent pixel correlation, and the easier it is for attackers to compromise. Unencrypted plaintext images typically exhibit strong correlations, whereas encrypted ciphertext images show low neighboring pixel correlations due to resetting the pixel values and positions. By testing horizontal, vertical, and diagonal correlations between the original image and its ciphertext across R, G, and B channels, the encryption efficacy can be evaluated.

In this section, using the Longmen image as an example, we calculate directional correlations (Table 4) and illustrate pixel distributions in R (blue), G (red), and B (yellow) channels (Fig. 23). Table 4 reveals that

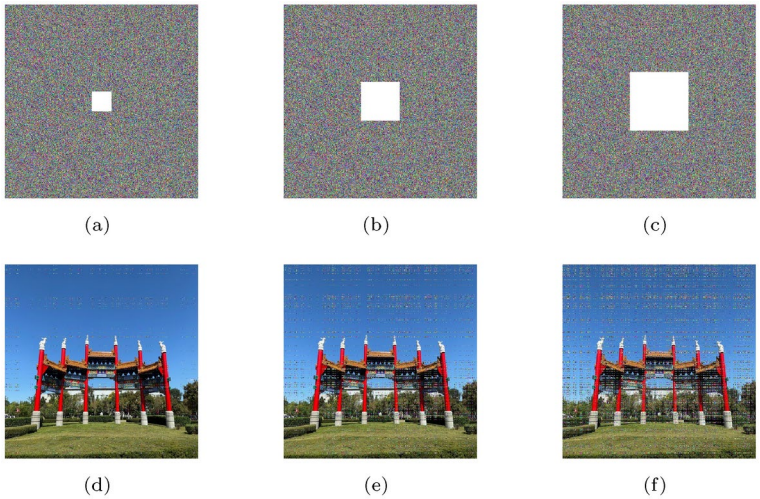


Fig. 22. Anti-cropping performance. (**a**) Cutting 0.1 Sipid image. (**b**) Cutting 0.2 Sipid image. (**c**) Cutting 0.3 Sipid image. (**d**) Cutting 0.1 decryption image. (**e**) Cutting 0.2 decryption image. (**f**) Cutting 0.3 decryption image.

Channel	Direction	Correlation (plaintext/ciphertext)
R-channel	Horizontal	0.9270/ 0.0054
	Vertical	0.9272/– 0.0176
	Diagonal	0.8739/– 0.0104
G-channel	Horizontal	0.9272/0.0036
	Vertical	0.9082/– 0.0061
	B-channel	0.8729/0.0031
B-channel	Horizontal	0.9742/– 0.0141
	Vertical	0.9684/– 0.0148
	Diagonal	0.9551/0.0020

Table 4. Comparison of information entropy between raw and ciphertext images.

the plaintext image's correlations approach 1, whereas the ciphertext image's values are near 0. Fig. 23 further demonstrates that plaintext pixels cluster along the diagonal, while ciphertext pixels disperse uniformly. These results confirm that the algorithm effectively conceals plaintext information.

Table 5 provides a comparative analysis between the proposed algorithm and those developed in prior studies. The results demonstrate that our algorithm achieves superior security performance and exhibits a significant competitive advantage over existing encryption methods.

Noise resistance performance

During the transmission of information over a channel, it may be interfered with by channel noise, leading to potential distortions in the received information. In this section, the Longmen image is used as an example. The first row in Fig. 24 depicts images corrupted by Gaussian noise with variances of 0.005, 0.01, and 0.015, respectively. It is evident that the visual quality of the image has deteriorated, yet details of the subjects remain discernible. The second row in Fig. 24 demonstrates decrypted images after salt-and-pepper noise (with densities of 0.1, 0.2, and 0.3) is applied to the ciphertext. The results indicate that salt-and-pepper noise has minimal impact on the proposed algorithm, with no significant degradation in image quality. These experiments verify the algorithm's robustness against noise interference.

Conclusion

In this paper, a 3D conservative chaotic system is successfully constructed through the introduction of a generalized Hamiltonian formulation and applied to amplitude modulation control of state variables. The system achieves flexible switching between chaotic and periodic states through parameter variations, while exhibiting a multi-stable coexistence phenomenon with rich dynamic behaviors. These results not only expand the applications of chaotic systems but also provide a new theoretical basis for controlling complex systems. At the application level, the proposed system is implemented in color image encryption algorithms, demonstrating excellent encryption performance and robust resistance against external attacks. Experimental results show that the algorithm exhibits superior performance in key space size, key sensitivity, and resistance against cropping

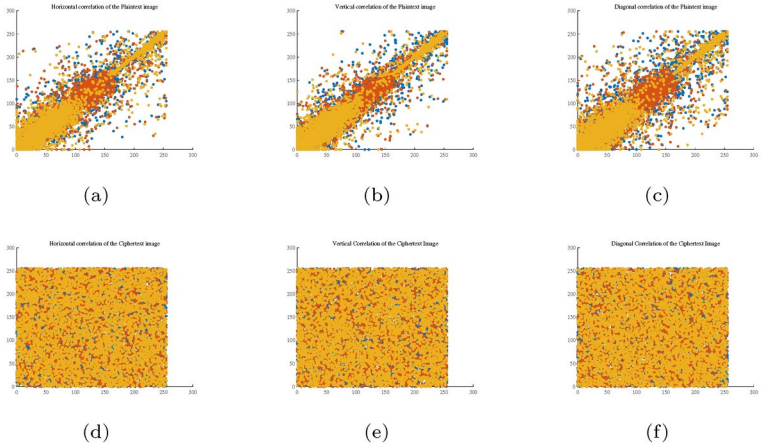


Fig. 23. Correlation analysis of Longmen image. (**a**) Horizontal orientation of the plaintext image. (**b**) Vertical orientation of the plaintext image. (**c**) Diagonal orientation of the plaintext image. (**d**) Horizontal orientation of the ciphertext image. (**e**) Vertical orientation of the ciphertext image. (**f**) Diagonal orientation of the ciphertext image.

Channel	Direction	This paper	Ref.[23]	Ref.[24]	Ref.[25]
R-channel	Horizontal	0.0158	0.0137	− 0.0209	− 0.0131
	Vertical	0.0139	− 0.0237	0.0489	0.0142
	Diagonal	− 0.0001	0.0109	0.0179	− 0.0044
G-channel	Horizontal	− 0.0071	− 0.0246	0.0081	− 0.0007
	Vertical	− 0.0063	− 0.0170	0.0108	− 0.0167
	B-channel	0.0156	− 0.0133	− 0.0125	− 0.0145
B-channel	Horizontal	− 0.0004	− 0.0137	− 0.0088	0.0036
	Vertical	− 0.0252	0.0023	0.0180	0.0083
	Diagonal	− 0.0007	− 0.0013	0.0349	− 0.0214
	Average	0.0101	0.0134	0.0201	0.0108

Table 5. Comparative analysis of correlation coefficients.

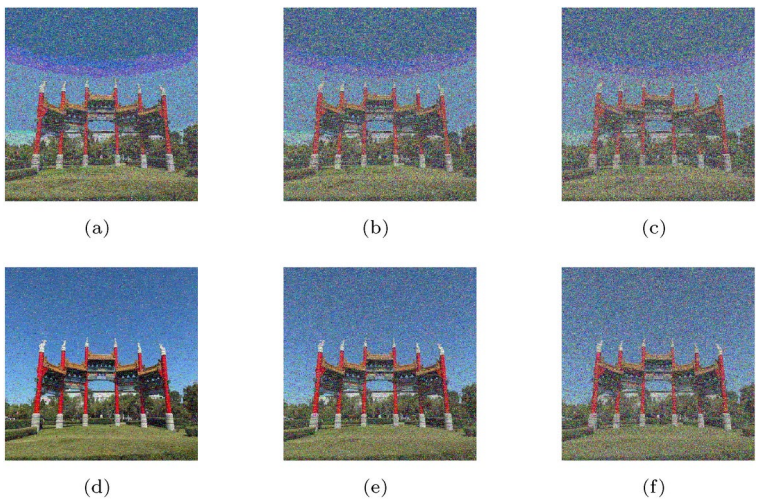


Fig. 24. Effect of adding Gaussian noise and salt-and-pepper noise on decryption. (**a**) Variance 0.005. (**b**) Variance 0.01. (**c**) Variance 0.015. (**d**) Noise density 0.1. (**e**) Noise density 0.2. (**f**) Noise density 0.3.

attacks, effectively resisting common cryptographic attacks. This paper provides a novel solution for image encryption, significantly enhancing the security and reliability of existing techniques. In conclusion, this paper not only advances the understanding of conservative chaotic systems but also opens new avenues for their practical engineering applications. Future work will explore the system's potential in other domains, such as secure communication and data encryption, to drive technological advancements in related fields.

Data availibility

All data generated or analysed during this study are included in this published article.

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Author contributions

Minxiu Yan: Writing – review & editing, Writing – original draft, Conceptualization. Minghui Liu: Writing – original draft, Software, Formal analysis. Chong Li: Validation, Project administration, Investigation. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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